Marked determinant charge and signed first-order cost

27 September 2026. Continues `fbe5404` and the I40–I49 proofs. The external reviews are preserved under `guidance/source/`; their proposed general no-loss theorem is a research target, not a premise here.

The new connection is between a marked **whole smooth core** and the actual determinant connection, without requiring a nonzero spinor inside the rescaled core. Its relative charge is controlled at every finite tree depth. This is not yet control of the opposite-spinor second Chern number: an explicit classical family distinguishes those charges. A separate signed localization below identifies the flux needed to improve the balanced small-h coefficient.

1. A marked determinant atom

Use the complex (physical) orientation of $\Gamma_n = \frac{1}{n}(1,-2)$, where $n = 2k+1 \geqslant 3$. The generator acts as $(z_1, z_2) \mapsto (\zeta z_1, \zeta^{-2} z_2)$. Put

$$f_n = \frac{k}{2k+1} = \frac{n-1}{2n} \in [1/3, 1/2). \tag{1.1}$$

Let Ω be a compact oriented smooth four-manifold with this single marked boundary. Let L be a Hermitian line bundle whose boundary flat character, in this marking, is ζ or ζ^{-1}. No topology or metric restriction is imposed on the interior. Write $F_A = i\omega_A$, with ω_A real, for a unitary connection.

1.1. The relative square and its sign

The minimal resolution has intersection matrix Q_k with diagonal $(-2, \dots, -2, -3)$ and adjacent off-diagonal entries 1. Its anticanonical class evaluates as $(0, \dots, 0, -1)$; its boundary character is ζ^{-1}. Inverting the last column gives

$$(Q_k^{-1})_{ik} = -\frac{i}{2k+1}, \qquad (Q_k^{-1})_{kk} = -f_n. \tag{1.2}$$

Indeed the interior recurrence is $x_{i-1} - 2x_i + x_{i+1} = 0$, with $x_0 = 0$ and $x_{k-1} - 3x_k = 1$. The inverse line has the same square. Glue an arbitrary same-marked filling to the reversed resolution, matching the flat boundary connections. The resulting closed line bundle has integral first-Chern square. Consequently, if A is flat on a boundary collar,

$$q_A := \frac{1}{4\pi^2} \int_\Omega \omega_A \wedge \omega_A \in \mathbb{Z} - f_n. \tag{1.3}$$

The statement uses the actual character, not just that it is faithful. A different character can have a different fractional square.

For a general boundary connection let A_0 be the flat connection on the same boundary line. Write $A = A_0 + ia$ there and set

$$b_A(Y) = \frac{1}{4\pi^2} \int_Y a \wedge da, \qquad q_A^{\mathrm{rel}} = q_A - b_A(Y). \tag{1.4}$$

These quantities are unambiguous: $H^1(Y;\mathbb{R}) = 0$, so changing the flat reference or gauge changes a by an exact form and does not change the integral. Flatten the connection near the boundary, or use the abelian transgression $d(a \wedge da) = da \wedge da$, to obtain

$$\boxed{q_A^{\mathrm{rel}} \in \mathbb{Z} - f_n, \qquad |q_A^{\mathrm{rel}}| \geqslant f_n.} \tag{1.5}$$

Boundary orientation is outward throughout. In particular, the flat-collar template satisfies

$$\boxed{\int_\Omega |F_A|^2 \geqslant 4\pi^2 f_n = 2\pi^2 \frac{n-1}{n} \geqslant \frac{4\pi^2}{3}.} \tag{1.6}$$

This is a characteristic-number energy bound, not an SW existence theorem. An actual positive-weight SW solution with identically flat determinant connection on an open collar would have zero spinor there. The application therefore uses the general-boundary estimate below, not a flat actual collar.

1.2. Arbitrary actual connection on a controlled geometric collar

Suppose an outer collar is identified with $[r, 2r] \times S^3/\Gamma_n$, with its metric uniformly bilipschitz to the Euclidean annulus and its radial coarea factors uniformly comparable. The comparison constant is fixed; r, n, interior geometry and internal scale separation are unrestricted.

For every smooth level Y_s, take the round Coulomb primitive of the restricted curvature. On the unit round quotient, the relevant positive Hodge spectrum is bounded below by that of S^3: lift to invariant forms, solve on S^3, and descend. Thus the constant is independent of n. Scaling and metric comparison give

$$\|a\|_{L^2(Y_s)} \leqslant Cr \|F_A|_{Y_s}\|_{L^2(Y_s)}, \qquad |b_A(Y_s)| \leqslant Cr \int_{Y_s} |F_A|^2. \tag{1.7}$$

Coarea supplies a level with

$$|b_A(Y_s)| \leqslant C \int_{\mathrm{collar}} |F_A|^2. \tag{1.8}$$

Let Ω_s be the whole region inside that level and let $\widehat{\Omega}$ include it and the collar. Combining (1.4), (1.8) and $|\omega_A \wedge \omega_A| \leqslant |F_A|^2 d\mu$ proves

$$\boxed{f_n \leqslant |q_A^{\mathrm{rel}}(\Omega_s)| \leqslant C \int_{\widehat{\Omega}} |F_A|^2.} \tag{1.9}$$

The relative number is independent of the selected level. No smallness of the connection, no gauge equation and no field-amplitude premise is required for (1.9). The geometric collar, marked character and whole filling are essential hypotheses. A terminal punctured region does not meet them.

1.3. Sharpness of the flat comparison template

Calderbank–Singer, Theorem A supplies scalar-flat Kähler ALE metrics on the minimal resolution. Their canonical determinant curvature is anti-self-dual in the physical orientation, has relative square $-f_n$, and hence has energy exactly $4\pi^2 f_n$. Truncate sufficiently far out and flatten the decaying connection to its flat reference over the outer annulus. The extra energy tends to zero. If desired, flatten the metric collar as in I47. Homothety does not change the four-dimensional curvature-square energy. Therefore

$$\inf_{\text{flat-collared marked smooth fillings}} \int |F_A|^2 = 4\pi^2 f_n. \tag{1.10}$$

This invokes a classical construction; it is not a new ALE metric or a closed actual zero-end escape sequence.

2. Actual determinant charge is tight on small-volume marked cores

Retain the actual finite-pencil input and class-square hypothesis of I41, Sections 6–7. For $\beta \geqslant 4$ and $0 < \eta \leqslant 1$, write

$$\eta = \kappa_\beta + \varepsilon, \qquad \epsilon_* = \beta^{-1} + \sqrt{\eta}.$$

The positive determinant fit and the negative determinant saturation give bounded two-form models K_+, K_- with

$$\|K_+\|_\infty + \|K_-\|_\infty \leqslant C, \qquad \int (|F_A^+ - K_+|^2 + |F_A^- - K_-|^2) \leqslant C\epsilon_* V. \tag{2.1}$$

Equivalently for the present purpose, there is a positive measure ν such that

$$|F_A|^2 d\mu \leqslant C\, d\mu + \nu, \qquad \nu(M) \leqslant C\epsilon_* V. \tag{2.2}$$

This follows pointwise from the squared distance to the bounded models. It is normalized-volume tightness, not local scale-invariant regularity.

Let $\widehat{\Omega}_i$ be any finite collection of the marked whole-core regions of Section 1.2 with overlap at most Λ. Put $Q_i = q_A^{\mathrm{rel}}(\Omega_i)$. Summing (1.9) and (2.2) yields

$$\boxed{\frac{1}{V}\sum_i |Q_i| \leqslant C\frac{\sum_i \mathrm{Vol}(\widehat{\Omega}_i)}{V} + C\Lambda\epsilon_*, \qquad \frac{N}{3} \leqslant \sum_i |Q_i|.} \tag{2.3}$$

In a zero-end sequence, bounded overlap and total core volume $o(V)$ therefore force the **absolute relative determinant-charge density**, not just the number density, to tend to zero. No nonzero rescaled spinor in a core is assumed. The actual line and its boundary marking are retained.

Alternatively suppose each enlarged core has volume at most a sufficiently small fixed v_*. The volume term in the unnormalized (2.3) is at most CNv_*, and can be absorbed using $N/3 \leqslant \sum_i |Q_i|$. Thus

$$N/V \leqslant C\Lambda\epsilon_*. \tag{2.4}$$

For **absolute** exclusion of every such core, one needs the corresponding absolute error tending to zero; for example, total core volume tending to zero and $\epsilon_* V \to 0$. Normalized estimates alone do not give that.

2.1. What survives an arbitrarily deep finite decomposition

For a region with one outer and several inner marked boundaries define the following quantity. Each inner boundary is oriented outward from the removed **child core**, opposite to its induced boundary orientation in the region:

$$Q_A(W) = \frac{1}{4\pi^2}\int_W \omega_A \wedge \omega_A - b_A(\Upsilon_{\mathrm{out}}) + \sum b_A(\Upsilon_{\mathrm{in}}). \tag{2.5}$$

Matched restrictions of the **same actual connection** give exact additivity when pieces are glued. Internal boundary terms cancel before absolute values are taken. A one-child flat neck can have zero charge and zero energy; do not charge it as if it were a smooth cap.

Apply (2.3) to a disjoint frontier of whole subtrees, not to every nested vertex. Its constant is independent of tree depth. A sequence of finite trees of unbounded depth therefore obeys the same frontier estimate. This is a proved depth-independent accounting result **given** those marked regions. It does not construct a bubble tree from weak integral input, nor license an infinite unsigned sum over overlapping ancestors.

More explicitly, set $\phi_e = b_e - f_{n_e}$ at every marked interface. A shell with outgoing edge e_0 and incoming edges $e_1, \dots, e_s$ satisfies

$$q_A(W) = m_W + \phi_{e_0} - \sum_{a=1}^{s} \phi_{e_a}, \qquad m_W \in \mathbb{Z}. \tag{2.6}$$

Sum this identity over any finite retained tree. Only its outer root and unresolved frontier remain. Equations (1.8), (2.2), and (2.3) bound the **absolute** sum of frontier b_e, f_{n_e}, and hence ϕ_e by $o(V)$ when the frontier whole cores have total volume $o(V)$. They also control integer charges: if $Q_i = m_i - f_{n_i}$, then $|m_i| \leqslant 2|Q_i|$. This supplies the limiting estimate beyond formal finite-tree cancellation.

2.2. The same relative charge in the actual harmonic class

Let $\mathscr{H}_A = 2\pi i a_g^{\mathrm{h}}$ be harmonic curvature on the **same integral determinant line**, including its torsion restriction, and let $J_+ = \|F_A^+\|_2^2 - 4\pi^2 p$. The supplied actual-field estimates give $J_+ \leqslant C\eta V$. Hodge orthogonality and Stokes imply

$$\boxed{\|F_A^{\pm} - \mathscr{H}_A^{\pm}\|_2^2 = J_+, \qquad \|F_A - \mathscr{H}_A\|_2^2 = 2J_+.} \tag{2.7}$$

Indeed the difference is exact, say $i\,d\alpha$, and $\int d\alpha \wedge d\alpha = 0$. Its two chiral energies agree. Combining (2.1) and (2.7) gives the same measure majorant and setwise tightness for $\mathscr{H}_A$ as for F_A.

If two connections on this line differ by $i\alpha$, the change of $\int \omega_{\ A} \wedge \omega_{\ A}$ is the boundary integral of $2\alpha \wedge \omega_{\ A} + \alpha \wedge d\alpha$. This is precisely the change of the boundary expression in (1.4). Therefore

$$q^{\mathrm{rel}}_{A_{\mathrm{h}}}(\Omega_i) = q^{\mathrm{rel}}_{A}(\Omega_i). \tag{2.8}$$

There is no estimate for the global primitive α, nor any use of a first eigenvalue bound. The equality is topological; the energy control comes from (2.7). Replacing the integral line by only its real cohomology vector would lose the torsion boundary marking and is not permitted.

2.3. Density in the original budget

Now assume $p \geqslant A > 0$ and $\chi(M)/A \leqslant B$. At small Ricci residual $r = \fint |\mathrm{Ric} + g|^2$, Chern–Gauss–Bonnet yields

$$\frac{8\pi^2\chi}{V} \geqslant \frac{(4 - 2\sqrt{r})^2}{24} - \frac{r}{2} = \frac{2}{3} - \frac{2}{3}\sqrt{r} - \frac{r}{3}. \tag{2.9}$$

Here $\fint (s + 4)^2 \leqslant 4r$ and $\fint |\mathrm{Ric}^{\circ}|^2 \leqslant r$. For $r \leqslant 1/4$, (2.9) gives $V/A \leqslant 32\pi^2 B$. Positive determinant saturation gives $p/V \to 2/(9\pi^2)$ at the endpoint. Consequently

$$\frac{9\pi^2}{2}(1 + o(1)) \leqslant \frac{V}{A} \leqslant 12\pi^2 B(1 + o(1)). \tag{2.10}$$

In particular (2.3) is also an $o(A)$ charge statement. The same Euler identity bounds the total W^- energy by $C(B)V$, hence the curvature energy of $E_- = S^- \otimes L$ by $C(B)V$. None of this gives local noncollapse or a quartic derivative bound for the torsion field.

3. The determinant charge does not control the torsion index

The distinction can be made exact. Now additionally assume a Spin-c structure on the entire filling with determinant L, extending the specified boundary lift. An arbitrary line bundle from Section 1 need not be such a determinant. In an exactly flat-conical comparison collar with the two allowed parallel boundary spinors, let J_+ be the relative second Chern number of S^+, and J_- that of $S^- \otimes L$, using these boundary sections. Write

$$q^{\mathrm{rel}}_{A} = m - f_n, \quad B_n = \frac{5 - n}{2} + f_n, \quad j = 2\chi(\Omega) + 3\tau(\Omega) - \frac{5 - n}{2}. \tag{3.1}$$

Both m, j are integers. The spinor characteristic identities, together with the outward cone Gauss–Bonnet boundary term $1/n$, give

$$\boxed{J_+ = \frac{m - j}{4}, \qquad J_- - J_+ = 2q^{\mathrm{rel}}_{A} + \chi(\Omega) - \frac{1}{n} = \chi(\Omega) - 1 + 2m.} \tag{3.2}$$

For the first formula, $4c_2(S^+) = c_1(L)^2 - (p_1 + 2e)$ and the integrated last parenthesis is $2\chi + 3\tau - B_n = j - f_n$. For the second, the exact characteristic-form identity is $c_2(S^- \otimes L) - c_2(S^+) = 2c_1(L)^2 + e$. The parallel boundary references have zero transgression. These derivations also specify the sign and the necessary congruence $m \equiv j \pmod 4$.

On the canonical minimal-resolution models used in (1.10),

$$\chi = k+1, \quad \tau = -k, \quad m = j = J_+ = 0, \qquad J_- = k = \frac{n-1}{2}, \qquad |q_A^{\mathrm{rel}}| = f_n < \frac{1}{2}. \tag{3.3}$$

Thus even a zero positive-spinor relative index and uniformly bounded determinant energy do not bound the opposite-spinor index. The latter grows without bound. Actual nonparallel boundary sections can additionally carry integer homotopy defects; they must be tracked rather than silently replaced by the flat references.

Consequently (2.3) cannot alone eliminate the order-V signed torsion flux forced by the actual zero-end identities. One needs actual control of that index or flux, a uniform charge-per-core bound, or a different direct PDE argument. The classical model proves this limitation of a proposed **charge comparison**, not existence of a closed zero-end counterexample.

3.1. An actual relative-index escape criterion

The existing zero-end identities give

$$\frac{c_2(E_-)[M]}{V} = \frac{1}{8\pi^2 V}\int |W^-|^2 + \frac{5}{12\pi^2} + O(\epsilon_*). \tag{3.4}$$

Suppose **disjoint** marked whole cores contain all zeros of the actual cubic torsion t_Φ, which is nonzero on their boundaries. Let ν_i be the relative second Chern index defined by this actual boundary section, not by a reference parallel spinor. Excision then gives $\sum_i \nu_i = c_2(E_-)[M]$. Bounded overlap alone would not justify this index identity.

If their total volume is $o(V)$, (2.3) implies, for every fixed L_0,

$$\frac{1}{V}\sum_{\nu_i^+ \leqslant L_0|Q_i|} \nu_i^+ \longrightarrow 0. \tag{3.5}$$

Thus macroscopic positive torsion index must occur at unbounded $\nu_i^+/|Q_i|$, or outside this proposed marked-core cover. Equation (3.3) shows why topology alone does not bound that ratio.

There is also an **actual local gauge** reason not to impose a universal index/charge comparison on arbitrary cores. In the existing hyperbolic ball construction of I41, Section 15.1, the determinant line is trivial, so Stokes gives exactly $q_A^{\mathrm{rel}} = 0$. The actual cubic torsion has relative index +1, and $\int |F_A|^2 = O(\varepsilon^4)$. Ricci and Weyl errors vanish on that ball. Its boundary winding and large outgoing derivative still have to be matched to an exterior. The normalized full form defect on the shrinking ball was not shown to tend to zero. This is a local obstruction to a charge-only argument, not a closed O10 counterexample.

3.2. Actual hyperbolic cores with unbounded torsion index

The preceding local obstruction extends beyond a single unit of charge. It can carry arbitrarily large index on the same ball topology, with a uniformly nonzero torsion field on a **round fixed-ratio collar**.

Proposition (high-index local actual fields). Fix a hyperbolic metric with sectional curvature $-1/3$ and a point. For every integer $d \geqslant 1$ and sufficiently small $\varepsilon > 0$, there is a smooth actual solution on the physical ball $B_{2\varepsilon}$, on its trivial determinant line, such that

$$D_A\Phi = 0, \qquad F_A^+ = 2i\sigma(\Phi), \qquad |\Phi|^2 = \frac{4}{3} + O_d(\varepsilon^2). \tag{3.6}$$

On every round sphere of radius $s\varepsilon$, $1 \leqslant s \leqslant 3/2$, its cubic torsion is nonzero and has relative index d^2. The relative determinant square is zero. Uniformly over these radii,

$$\int_{Y_{s\varepsilon}} G = -d^2 + O_d(\varepsilon^4), \qquad q_A^{\mathrm{rel}} = 0. \tag{3.7}$$

Here G is the fixed torsion-cutoff transgression of Section 5, which equals the full unit-section transgression throughout that collar.

Construction. Use the Clifford frame and curved right inverses of I41 Section 15.1. In rescaled coordinates $x = \varepsilon y$, set

$$a^2 = 4/3, \quad c_d = 2^{(d-1)/2}, \quad g_d(z) = \frac{c_d}{2(d+1)}(\bar z_1^{d+1} - \bar z_2^{d+1}), \quad \Phi_0 = ae_1 + \varepsilon g_d e_2. \tag{3.8}$$

The antiholomorphic coefficient has $D_0^+(0, g_d) = 0$. Direct substitution in the cubic torsion formula gives, in physical derivative normalization,

$$t_0(\varepsilon y) = a^2 c_d(\bar z_1^d, \bar z_2^d). \tag{3.9}$$

Indeed the components are $a^2(g_1 + ig_2, -g_3 - ig_4)$; the terms involving εg_d cancel by the same antiholomorphic Dirac equations.

For each fixed d, the pulled-back rescaled metric is Euclidean plus $O(\varepsilon^2)$ in every fixed differentiability norm. The Dirichlet right inverses R_D, R_A used in the earlier construction are uniformly bounded for small ε. First correct the seed by $-R_D D_\varepsilon^+ \Phi_0 = O_d(\varepsilon^2)$. Then apply the contraction

$$A(\Phi) = R_A(0, 2\sqrt{2}\, i\varepsilon^2\sigma(\Phi)), \qquad \Phi = \widehat\Phi_0 - \tfrac{1}{2} R_D[c(A(\Phi))\Phi]. \tag{3.10}$$

On a fixed $C^{k,\alpha}$ neighborhood, $k \geqslant 3$, its Lipschitz constant is $O_d(\varepsilon^2)$. It gives a smooth exact solution with $A_y = O_d(\varepsilon^2)$ and $\Phi - \Phi_0 = O_d(\varepsilon^2)$, and hence

$$t(\varepsilon y) = a^2 c_d(\bar z_1^d, \bar z_2^d) + O_d(\varepsilon) \quad \text{in } C^1. \tag{3.11}$$

Constants may depend on d. Boundary values are outputs of the construction; they have not been prescribed as gluing data.

Since $c_d^2(|z_1|^{2d} + |z_2|^{2d}) \geqslant |y|^{2d}$, the actual torsion has $|t| \geqslant a^2/2$ on $1 \leqslant |y| \leqslant 3/2$, for sufficiently small ε. Its squared norm is at least $4/9$, exceeding the cutoff threshold $m_\beta/8 \leqslant 16/81$. Thus $\theta = 1$ on that whole collar, for every $\beta \geqslant 4$.

The map $(\bar{z}_1^d, \bar{z}_2^d)$ has real degree $+d^2$: the holomorphic power map has degree d^2, and conjugation on two complex coordinates preserves real orientation. Boundary nonvanishing and the small perturbation preserve this degree on every indicated sphere. Interior zeros may split or be nontransverse; no unique-zero assertion is needed for the relative index. Curvature of E_- is $O_d(1)$ in physical norms, so $\int_{B_{s\varepsilon}} c_2(E_-) = O_d(\varepsilon^4)$. The relative Chern formula proves (3.7). The determinant equality in (3.7) is exact Stokes on a trivial line, independently of any norm estimate.

The construction and exact torsion equation also give

$$\begin{aligned} \int (|\nabla_A^2 \Phi|^2 + |\nabla t|^2) &= O_d(\varepsilon^2), \\ \int (|F_A|^2 + |\nabla_A \Phi|^4 + |t|^4) &= O_d(\varepsilon^4), \\ \int (|\langle \Phi, \nabla_A \Phi \rangle|^2 + |D_{3A}^- t|^2) &= O_d(\varepsilon^6). \end{aligned} \tag{3.12}$$

For the last estimate, the current is $O_d(\varepsilon)$, the spinor derivative is $O_d(1)$, and the actual torsion Dirac identity cancels the scalar/model terms; $W^+ = 0$ here. Ricci and Weyl errors vanish exactly because the physical metric is the fixed hyperbolic metric.

Choose ε_d diagonally, after fixing d, so all costs in (3.12) tend to zero and the boundary degree and cutoff estimates hold. Then $\nu = d^2 \to \infty$, while the displayed absolute subcritical costs vanish. There is no claim of a uniform-in-d implicit-function radius.

The missing critical direction cost is visible explicitly. For the unit seed $u_d = t_0/|t_0|$, polar coordinates give

$$\frac{d^2}{r^2} \leqslant |du_d|^2 \leqslant \frac{4d^2}{r^2} \quad (1 \leqslant r \leqslant 3/2). \tag{3.13}$$

The phase contribution is $d^2(x^{d-1} + y^{d-1})/(r^2(x^d + y^d))$, where $x = \cos^2 \vartheta$, $y = \sin^2 \vartheta$. It lies between d^2/r^2 and $2d^2/r^2$. The remaining angular derivative is bounded by d^2/r^2: its ratio to this bound is $(xy)^{d-1}/(x^d + y^d)^2 \leqslant 1$. For example write $x/y = z^2$; the last inequality follows from $(z^{d-1} - 1)(z^{d+1} - 1) \geqslant 0$, using $x + y = 1$. The phase upper bound follows from $(x - y)(x^{d-1} - y^{d-1}) \geqslant 0$. Endpoint values follow by continuity. The displayed upper bound is deliberately relaxed. For each fixed d, (3.11) transfers these estimates to the actual unit field after decreasing ε. The diagonal can therefore retain

$$\int_{\varepsilon_d \leqslant r \leqslant 3\varepsilon_d/2} |\nabla u|^4 \asymp d^4 \longrightarrow \infty, \quad \int_{\varepsilon_d \leqslant r \leqslant 3\varepsilon_d/2} |\nabla u|^2 \asymp \varepsilon_d^2 d^2 \longrightarrow 0. \tag{3.14}$$

This disproves a local signed-flux bound based only on the displayed subcritical budgets, even with actual gauge equations and a nonzero fixed-ratio torsion collar. It does **not** supply a fully calibrated endpoint collar: torsion length has large radial/angular variation, and the ASD determinant fit to the required exterior fails. There is no closed conformal supplier, global class-square budget, or exterior matching. It is not an O10 counterexample, nor a sharpness example for the frontier exponent in (5.5).

4. Where a smaller balanced first-order coefficient must be paid

This section concerns the established **balanced surgery family**, not the zero-end weighted solution. Let

$$M = X\#k\overline{\mathbb{CP}}^2\#\ell(S^1\times S^3),\quad A = c_1^2(X),\quad h = (k+4\ell)/A \to 0,\quad U = 1+h+O(h^2). \tag{4.1}$$

Use a scalar-minus-four metric and one actual **ordinary**, unweighted SW solution (A_{conn},Φ) supplied by the specified class with $a^2 = A-k$ and $p = (a_g^+)^2 \geqslant A$. The symbol A without a subscript continues to denote the topological budget, not the connection. In the formulas below $F_{A_{\mathrm{conn}}}$ is abbreviated to F_A. All fields and remainders in this section belong to this same solution.

Put

$$\rho = |\Phi|^2 \leqslant 4,\quad P = \int |\operatorname{Ric}^\circ|^2,\quad R = P+4V,\quad \mathscr{E} = 4P/R,\quad J = \|F_A^+ - (F_A^+)_H\|_2^2.$$

The actual input and Hodge identities give

$$c_2(S^+)[M] = \ell,\qquad B := \int |F_A^-|^2 = 4\pi^2(p-A+k)+J. \tag{4.2}$$

Choose a fixed smooth increasing $v(\rho)$ equal to zero for $\rho \leqslant 1/2$, one for $\rho \geqslant 1$, with $|v'| \leqslant 4$, and put $\chi = (1-v)^2$. Write T_u for the exact rank-two transgression on $u = \Phi/\sqrt{\rho}$, as in I40 Section 1, and define

$$q_\chi = \int d\chi\wedge T_u,\quad P_\chi = \int \chi|\operatorname{Ric}^\circ|^2,\quad P_o = P-P_\chi,\quad B_\chi = \int \chi|F_A^-|^2,\quad B_o = B-B_\chi,$$
$$M_\chi = \int \chi\left(2|W^+|^2+\frac{2}{3}-\frac{\rho^2}{8}\right) \geqslant \frac{13}{24}\int\chi \geqslant 0. \tag{4.3}$$

The signed flux q_χ is a different object from the relative determinant square q_A^{rel}. In particular, the low-amplitude region has not been identified as a marked whole surgery core.

4.1. Exact spatial balances

The transgression identity $dT_u = c_2(\mathbb{F})$ and the extension of $(1-\chi)T_u$ by zero near all spinor zeros give

$$\ell = \int \chi c_2(\mathbb{F}) + q_\chi.$$

There is no transversality or regular-level-set assumption. For the actual positive-spinor connection the signed Chern density is

$$16\pi^2 c_2(\mathbb{F}) = \left(|F_A^+|^2 - |F_A^-|^2 - 2|W^+|^2 - \frac{s^2}{24} + \frac{|\operatorname{Ric}^\circ|^2}{2}\right)d\mu.$$

Use $s = -4$ and $|F_A^+|^2 = \rho^2/8$. This proves the two exact identities

$$\boxed{P_\chi = 32\pi^2\ell + 2B_\chi + 2M_\chi - 32\pi^2 q_\chi,} \tag{4.4}$$

$$\boxed{P - 8\pi^2 Ah = P_o + 8\pi^2(p - A) + 2J - 2B_o + 2M_\chi - 32\pi^2 q_\chi.} \tag{4.5}$$

For (4.5), substitute (4.2) in (4.4) and use $Ah = k + 4\ell$. The only unfavorable terms are exterior ASD determinant energy and positive signed transition flux. This specifies a spatial source of the earlier signed remainder; it does not discard the Weyl payment tested by X06.

4.2. First-order necessary conditions

Suppose $\mathscr{E} = O(h)$. With $e = R/(8\pi^2 A)$, the two classical energy walls give

$$1 + h \leqslant e \leqslant \frac{U}{1 - \mathscr{E}} = 1 + O(h).$$

Thus on any subsequence on which $\mathscr{E}/h \to c$, $P/(Ah) \to 2\pi^2 c$. Moreover $U = 1 + 15k/A - 12\ell/A$ implies

$$\frac{k}{Ah} \to \frac{2}{9}, \qquad \frac{\ell}{Ah} \to \frac{7}{36}. \tag{4.6}$$

Dropping only the displayed nonnegative terms in (4.4) gives

$$\boxed{\liminf \frac{q_\chi}{Ah} \geqslant \frac{7}{36} - \frac{c}{16}.} \tag{4.7}$$

In particular, a coefficient below 28/9 requires positive signed flux of first-order size. Conversely, the **additional** condition $\limsup q_\chi/(Ah) \leqslant 0$ forces $c \geqslant 28/9$, not four. The class supply does not establish that sign condition.

For every $c < 4$, (4.5) instead gives

$$\boxed{\liminf \frac{B_o + 16\pi^2 q_\chi}{Ah} \geqslant \pi^2(4 - c).} \tag{4.8}$$

Thus proving both $B_o = o(Ah)$ and $(q_\chi)_+ = o(Ah)$ would suffice for the coefficient-four lower bound. These are substantive missing estimates, not consequences asserted here of near-minimality.

4.3. The flux demands an actual critical mixed current

Set $T = \{1/2 < \rho < 1\}$, $\zeta = \langle \Phi, \nabla_A \Phi \rangle$, and

$$M_{\mathrm{mix}} = \int_T |\zeta|^2 |\nabla_A \Phi|^2.$$

For this ordinary solution, the exact nonnegative remainder in I47, Section 4.1 reads

$$D_2/18 = F_{\mathrm{fit}} + \Gamma/2 + J + H,$$

$$\Gamma = \int \left(\frac{|d\rho|^2}{4} + |\operatorname{Im}\zeta|^2 \right), \quad H = \frac{2}{3}\int (1-x)^2(2+4x-x^2), \quad x = \rho/4,$$

$$F_{\mathrm{fit}} = \int |W^+ - W_0|^2, \quad W_0 = \tfrac{1}{2}(|\sigma(\Phi)|^2 I/3 - \sigma(\Phi)\otimes\sigma(\Phi)). \tag{4.9}$$

In particular $Q := \int (4-\rho)^2 \leqslant 2D_2/3$ and $\Gamma \leqslant D_2/9$. On T, volume is at most $Q/9$, the model W_0 is bounded, and $F_{\mathrm{fit}} \leqslant D_2/18$. If Ω denotes trace-free positive-spinor curvature, then $G^2 = |\Omega|^2 + |F_A^+|^2$ satisfies

$$\int_T G^2 \leqslant C(P + D_2).$$

The exact transgression in a unitary (u, v) frame is

$$4\pi^2 T_u = -a_{\mathrm{conn}} \wedge \mathbb{F}_{22} + \tfrac{1}{2}(b \wedge \mathbb{F}_{12} - \bar{b} \wedge \mathbb{F}_{21}) + a_{\mathrm{conn}} \wedge \bar{b} \wedge b.$$

On T, $|d\rho| \leqslant 2|\zeta|$, $|a_{\mathrm{conn}}| \leqslant 2|\zeta|$, and $|b| \leqslant \sqrt{2}|\nabla_A \Phi|$. Hence Cauchy–Schwarz, without a Sobolev or Hessian estimate, gives

$$\boxed{|q_\chi| \leqslant C\left\{\sqrt{(P + D_2 + B)M_{\mathrm{mix}}} + M_{\mathrm{mix}}\right\}.} \tag{4.10}$$

The central term is controlled using $\int_T |\zeta|^4 \leqslant M_{\mathrm{mix}}$. Along the above first-order sequences,

$$D_2 = 36\pi^2 A\{e - 1 - h - 2(p/A - 1)\} \geqslant 0$$

implies $D_2, p - A = O(Ah)$, and (4.2), (4.9) give $B = O(Ah)$. Thus a coefficient bounded strictly below $28/9$ forces $\liminf M_{\mathrm{mix}}/(Ah) > 0$. This is a necessary actual concentration, not its exclusion.

If additionally $D_2 = o(Ah)$, then $\mathrm{Vol}(T), \Gamma = o(Ah)$. To control the ordinary spinor's quadratic derivative there, test $4\nabla_A^*\nabla_A\Phi + (\rho - 4)\Phi = 0$ against $\theta(\rho)\Phi$, where $\theta = 1$ below 1, zero above 2, and $\theta' \leqslant 0$ is bounded. Integration by parts gives

$$4\int \theta |\nabla_A \Phi|^2 = \int \theta(4-\rho)\rho - 2\int \theta' |d\rho|^2 \leqslant C(Q + \Gamma) \leqslant C D_2. \tag{4.11}$$

Therefore below the stated threshold,

$$\mathrm{Vol}(T), \ \int_T |\nabla_A \Phi|^2 = o(Ah), \qquad \int_T |\zeta|^2 |\nabla_A \Phi|^2 \geqslant c_* Ah. \tag{4.12}$$

In particular the derivative supremum diverges, since $M_{\mathrm{mix}} \leqslant \sup_T |\nabla_A \Phi|^2 \, \Gamma$. No estimate for a different weighted solution was substituted here.

4.4. Conditional critical patterns

If $c = 28/9$ and $\limsup q_\chi/(Ah) \leqslant 0$, equality in the limiting nonnegative balance (4.4) forces

$$P_o, B_\chi, M_\chi = o(Ah), \quad q_\chi = o(Ah), \quad \liminf \frac{B_o}{Ah} \geqslant \frac{8\pi^2}{9}. \tag{4.13}$$

All leading Ricci error lies in the low-amplitude region, while the indicated ASD determinant energy lies outside. Since $\int \chi = o(Ah)$ but $P_\chi/(Ah) \to 56\pi^2/9$, $\|\operatorname{Ric}^\circ\|_\infty$ diverges on that region.

If instead $c = 4$, $B_o = o(Ah)$, and $(q_\chi)_+ = o(Ah)$, then (4.5) forces

$$P_o, p - A, J, M_\chi, q_\chi = o(Ah), \quad \frac{B_\chi}{Ah} \to \frac{8\pi^2}{9}, \quad \frac{P_\chi}{Ah} \to 8\pi^2. \tag{4.14}$$

These are different conditional first-order patterns. Neither is a classification of all minimizing sequences, and neither proves that the coefficient-four sufficient conditions hold on the classical recovery family.

5. A quantitative criterion for torsion flux at a marked frontier

Return to the weighted zero-end solution. Let $t = t_\Phi$, $m = |t|^2$, and use the inherited torsion cutoff $\chi = 1 - \theta(m)^2$, with $\theta = 0$ for $m \leqslant m_\beta/16$ and $\theta = 1$ for $m \geqslant m_\beta/8$. Here m_β has positive upper and lower bounds for $\beta \geqslant 4$. Define $G = \theta^2 T_t$, extended by zero through torsion zeros. This χ is the torsion cutoff, not the ordinary-spinor cutoff of Section 4.

With

$$k_A = 9(|F_A^+|^2 - |F_A^-|^2) + s^2/24 - |\operatorname{Ric}^\circ|^2/2,$$

the exact local version of the inherited signed identity is, for every smooth domain D,

$$2\int_D \theta^2 |W^-|^2 + \int_D \theta^2 k_A = 16\pi^2 \left(\int_D d\chi \wedge T_t + \int_{\partial D} G \right). \tag{5.1}$$

This follows by integrating $dG = -d\chi \wedge T_t + \theta^2 c_2(E_-)$. Boundary terms cancel for an actual partition, but need not disappear in a limiting decomposition.

5.1. A scale-free trace bound

Consider a controlled collar of radius r_e and order n_e as in Section 1.2. Put

$$E_e = \int_{\mathscr{C}_e} |F_{E_-}|^2, \qquad J_e = \int_{\mathscr{C}_e \cap \{m > m_\beta/16\}} |\nabla u|^4, \qquad u = t/|t|.$$

Here the derivative is that of the actual unit section, including its phase, not just its image in the projective bundle. The rank-two transgression gives

$$|G| \leqslant C\theta^2 (|F_{E_-}||\nabla u| + |\nabla u|^3).$$

Use coarea to choose an interior slice, then Hölder on the collar. Since $\mathrm{Vol}(\mathscr{C}_e) \leqslant C r_e^4/n_e$,

$$\boxed{\left|\int_{Y_e} G\right| \leqslant C n_e^{-1/4}(E_e^{1/2} J_e^{1/4} + J_e^{3/4}).} \tag{5.2}$$

The physical radius cancels. By selecting from the good sets of the two coarea estimates, the same slice can satisfy (1.8) and (5.2). Let $H = \sum_e n_e^{-1}$. Hölder in the interface index gives

$$\boxed{\frac{1}{V}\sum_e \left|\int_{Y_e} G\right| \leqslant C\left(\frac{H}{V}\right)^{1/4}\left[\left(\frac{\sum E_e}{V}\right)^{1/2}\left(\frac{\sum J_e}{V}\right)^{1/4} + \left(\frac{\sum J_e}{V}\right)^{3/4}\right].} \tag{5.3}$$

For marked frontiers of Section 2, $H/V \leqslant N/V \to 0$. If their normalized quartic torsion-direction energy stays bounded and their normalized curvature energy stays bounded, no normalized torsion boundary flux is lost at the selected cuts.

Under $p \geqslant A > 0$ and $\chi(M)/A \leqslant B$, the latter curvature bound follows from the original actual budgets, rather than an additional C^2 collar condition. In the project's curvature norms,

$$|F_{E_-}|^2 = |W^-|^2 + s^2/48 + |\,\mathrm{Ric}^\circ\,|^2/4 + (9/2)|F_A|^2.$$

Gauss–Bonnet gives $\int |W^-|^2 \leqslant 8\pi^2\chi + P/2$, and Section 2.3 controls χ/V. For bounded-overlap collars, therefore, $\sum E_e/V \leqslant C(B)\Lambda$.

5.2. Surviving flux forces a faster critical concentration

Write $d = H/V$, $j_4 = \sum J_e/V$, and suppose the left side of (5.3) is at least a fixed $b > 0$, while $\sum E_e/V \leqslant E_0$. At least one of its two summands must be at least $b/2$. Hence

$$j_4 \geqslant c \min\left\{\frac{b^4}{dE_0^2}, b^{4/3} d^{-1/3}\right\}. \tag{5.4}$$

For $d \to 0$ and fixed b, E_0, this implies

$$\boxed{j_4 \geqslant c(b, E_0) d^{-1/3}.} \tag{5.5}$$

In the uniformly small-core-volume version (2.4), $d \leqslant C\epsilon_*$, so surviving macroscopic flux forces $j_4 \geqslant c\epsilon_*^{-1/3}$. This is stronger than a merely positive quartic concentration requirement.

The quantifier matters: the macroscopic flux must survive the **coarea-selected cuts**, or every admissible cut, not just a preselected hypersurface. Moving a cut can transfer nonclosed G into the bulk and transition terms of (5.1). The global signed identity does not alone force all of its flux through a chosen marked frontier.

The lower bound for $|t|$ on this region implies $|\nabla u| \leqslant C|\nabla t|$. Current actual budgets therefore bound the normalized quadratic direction energy, but not the quartic quantity in (5.5). Pure radial amplitude change cannot by itself supply this boundary flux.

For completeness put $L_2 = V^{-1}\sum \int |\nabla u|^2$ over these collar regions. All sets below are restricted to $\bigcup_e(\mathscr{C}_e \cap \{m > m_\beta/16\})$; multiplicity is $\sum_e \mathbf{1}_{\mathscr{C}_e\cap\{m>m_\beta/16\}}$. With $L_2 \leqslant C\Lambda$ and $j_4 \to \infty$, the set $Z = \{|\nabla u|^2 \geqslant j_4/(2L_2)\}$, counted with this multiplicity, satisfies

$$\frac{\mathrm{Vol}_{\mathrm{mult}}(Z)}{V} \leqslant \frac{2L_2^2}{j_4}, \qquad \frac{1}{V}\sum_e \int_{\mathscr{C}_e\cap Z} |\nabla u|^4 \geqslant j_4/2. \tag{5.6}$$

Thus the forced divergent quartic mass lies on vanishing relative volume. Equations (5.5)–(5.6) in particular force $\sup|\nabla u| \geqslant cd^{-1/6}$, and hence a corresponding bound for the physical $|\nabla t|$ on that region.

5.3. An explicit routing condition, and the remaining alternative

Here is one sufficient condition that ensures the required macroscopic flux survives the selected cuts. Suppose disjoint marked whole cores have total volume $o(V)$, all the torsion low-amplitude region $\{\chi \neq 0\}$ lies strictly inside them, and $\chi = 0$ throughout every admissible outer collar and the exterior. This is an additional actual-field condition.

For any coarea-selected cuts, apply (5.1) to their exterior D. Its transition term is zero. The inherited fits give $V^{-1}\int_M k_A = 20/3 + O(\epsilon_*)$ and $V^{-1}\int_{M\setminus D}|k_A| = o(1)$. Since the W^- term is nonnegative, the sum of **outward core** fluxes is negative and obeys

$$-\frac{1}{V}\sum_e \int_{Y_e} G \geqslant \frac{5}{12\pi^2} - o(1). \tag{5.7}$$

If also $p \geqslant A > 0$ and $\chi(M)/A \leqslant B$, the curvature input to (5.5) is automatic. The resulting alternative is precise: such a routing decomposition of an actual endpoint sequence must have

$$\frac{1}{V}\sum_e \int_{\mathscr{C}_e\cap\{m>m_\beta/16\}} |\nabla u|^4 \geqslant c(B,K)(H/V)^{-1/3} \longrightarrow \infty. \tag{5.8}$$

In particular a decomposition satisfying the strictly weaker-than-bounded condition $j_4 = o((H/V)^{-1/3})$ is impossible. This connects a genuine global field-routing premise to a quantified failure, rather than assuming an arbitrary boundary has macroscopic flux.

Weak input has not yet supplied this decomposition or ruled out (5.8). Trivial spherical interfaces are a further distinct possibility: (5.2) holds for them with $n = 1$, but the determinant atom does not bound their number. The actual local example in Section 3.1 explains why they cannot be silently omitted from a proposed decomposition. These alternatives leave the unrestricted endpoint theorem open.

5.4. Why level cuts alone do not cancel the signed trace

The phase part of the same transgression identifies a possible false shortcut. On $t \neq 0$, put $j_t = \mathrm{Im}\langle t, \nabla t\rangle$, $m = |t|^2$, and $f_A = -iF_A$. In a unit t-frame the diagonal connection is ij_t/m, and the determinant of E_- has curvature $3if_A$. The simplified rank-two formula gives exactly

$$4\pi^2 T_u = -m^{-2} j_t \wedge dj_t + 3m^{-1} j_t \wedge f_A + \frac{1}{2}(b \wedge \mathbb{F}_{12} - \bar{b} \wedge \mathbb{F}_{21}). \tag{5.9}$$

The explicit radial derivative cancels from the first term because $j_t \wedge dm \wedge j_t = 0$; the phase helicity does not cancel. Thus taking a mass-level boundary, where tangential $dm = 0$, does not force the trace to be small. The inherited hyperbolic local actual core already has outward $\int T_u = -1 + O(\varepsilon^4)$ on its nonzero outer section, while its bulk characteristic integral tends to zero. No nonpositive-flux theorem or improved signed trace bound follows from level selection alone.

6. Consequences for the continuing research

There are now two distinct actual connections, not one interchangeable charge:

- The determinant's full relative charge is tight on existing marked small whole cores, including harmonic replacement and arbitrary finite depth.
- Torsion indices can grow relative to that charge, and an actual local trivial-boundary core can carry torsion index with zero determinant charge. Any global argument must retain the actual section framing or directly control the resulting signed flux.

For the zero-end target, the next sufficient step is either direct actual control of the signed transition in (5.1), or a decomposition that proves both a genuine flux-routing statement and enough control to contradict (5.5). Proving the first does not require completing all possible core classifications. For balanced first-order sharpness, (4.5) instead identifies the two actual quantities whose leading contribution must be decided.

The present arguments do not give an unrestricted endpoint gap, a Gamma-limit, the balanced optimal coefficient, diameter-free realization, or the Einstein classification of the fixed Dehn-twist manifold. Those remain separate open statements. The written proofs above provide specific new conditional connections and identify exactly which proposed stronger charge comparisons are false.